Taking the Converse

Neal

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1 Abstract

This was based on some random conversations I had with Haruka Kimura.

Now the proper, 'official' name of this concept is called phantom points. But this is not how most geo mains I know think of it. Instead, it is literally taking the converse. To justify that the phantom object is in fact equal to the original object, we note a bunch of properties and then explain how those uniquely define something.

Motivating example (ISL 2025/G3)

Let ABC be an acute triangle with $AB < AC$. Let Γ be the circumcircle of ABC. The perpendicular from B to AC intersects Γ again at $D \neq B$. Let Q be the point on Γ, different from B, such that $AB = AQ$. Lines BQ and AC intersect at R. Line DR intersects Γ again at $S \neq D$. Segment BC intersects the circumcircle of triangle QRS at T. Lines TQ and RS intersect at X. Prove that CX is perpendicular to AB.

This problem derives a lot of difficulty from the contrived wording. In reality, line QTX could be more naturally defined via reflection of orthocenter in $\overline{BC}$. After realizing this, the problem is not very hard.

Credits. Tiger Zhang actually introduced this idea to me. See his solution of ISL 2021 G8 on the AoPS thread.

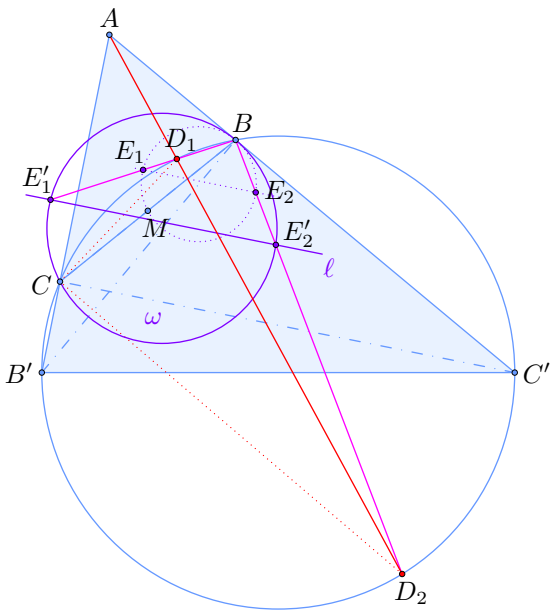
But this sounds much more like proving the converse of a probelm (or a claim/lemma within the geo) and then explaining why they are equivalent. But having done so many geo problems, I realized that most geometry problems are equivalent to their converses anyway. Though some problems are isomorphic to their converses, most problems do not have such extreme symmetry and it is a common trope to artificially increase a problem's difficulty (i.e. no real ideas were introduced).

2 Lecture

Since this is a short lecture I am too lazy to add exercises.

Example (USAMO 2024/5)

Point D is selected inside acute $\triangle ABC$ so that $\angle DAC = \angle ACB$ and $\angle BDC = 90^\circ + \angle BAC$. Point E is chosen on ray BD so that $AE = EC$. Let M be the midpoint of BC. Show that line AB is tangent to the circumcircle of triangle BEM.



Convert everything to direct angles.

- First, look at how D is defined. The first angle condition implies that D on a specific line thru A while the second implies that D lies on a specific circle through B, C.
- Those two curves intersect twice…so we can replace D with D_1, D_2.
- We can also just define the circle from the E's, as the perpendicular bisector of $\overline{AC}$ intersects the circle thru M tangent to $\overline{AB}$ at B twice.

Converse of problem

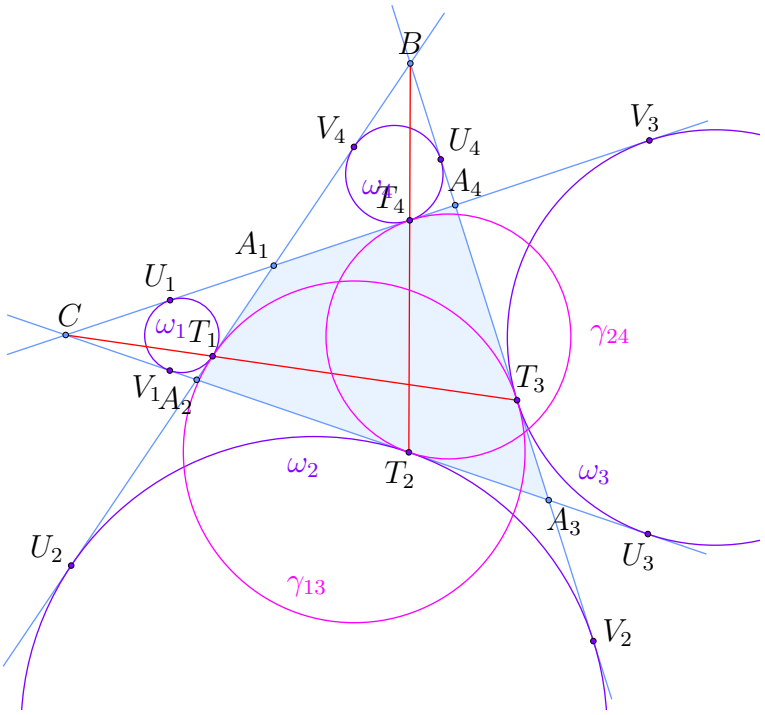
In acute triangle $AB'C'$, points B, C are the feet of the altitudes from B', C'. Let ℓ be the perpendicular bisector of $\overline{AC}$ scaled at B by 2, and circle ω be tangent to $\overline{AB}$ and through B, C, and suppose they meet at E_1', E_2'. Define $D_k = \overline{BE_k'} \cap (BC)$ for $k = 1, 2$. Then line D_1D_2 satisfies:

1. $\measuredangle(\overline{D_1D_2}, \overline{AC}) = \measuredangle ACB$;
2. It passes through A.

The rest of the problem is just using the elem construction principle to pin down what the line D_1D_2 has to be. I actually wrote 'take the converse' on the contest, and it received full credit!
Conventionally, one would say I used phantom lines but 'converse' felt more natural.

Example (RMM 2010/3)

Let $A_1A_2A_3A_4$ be a quadrilateral with no pair of parallel sides. For each $i = 1, 2, 3, 4$, define ω_i to be the circle touching the quadrilateral externally, and which is tangent to the lines $A_{i-1}A_i$, A_iA_{i+1} and $A_{i+1}A_{i+2}$ (indices considered modulo 4). Let T_i be the point of tangency of ω_i with the side A_iA_{i+1}. Prove that the lines A_1A_2, A_3A_4 and T_2T_4 are concurrent if and only if the lines A_2A_3, A_4A_1 and T_1T_3 are concurrent.



- After drawing in the four excircles/incircles, the concurrencies in the problem statement looked awfully like Monge. Indeed, if there existed a circle touching opposite sides at T_1, T_3, then it would fit Monge and the 3 lines would concur.
- Thus, the concurrence boils down to the existence of a circle. How do you show there exists a circle tangent to opposite sides like this?
- The natural choice is to intersect $X_1 = \overline{A_1A_2} \cap \overline{A_3A_4}$, then use the length condition $X_1T_1 = X_1T_3$. If we define X_2 similarly, it becomes apparent that we just want $X_1T_1 = X_1T_3$ to be equivalent to $X_2T_2 = X_2T_4$!
- We do need to add in more tangency points of the existing circles, but it's literally just adding and subtracting lengths because how else would I conclude the problem?

Here the problem uses another geo heuristic: a lot of problems of the form $A \iff B$ where A and B are symmetric variants can be solved by finding a third intermediary C with $A \iff C$ and $C \iff B$.
This problem, to use the 'phantom' language, would involve a phantom circle.

3 Conclusion

The idea of converses helps us generalize phantoms from not just redefining points.